Band-1 Computational Experiment: Graceful Tree Conjecture Finite Verification

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note is a **finite verification probe** (collapse list). It enumerates nonisomorphic trees on $n \leq N$ vertices and searches for graceful labelings. It does **not** prove the graceful tree conjecture, does **not** claim Clay Millennium status, and does **not** claim Theory of Everything progress. **Problem status: OPEN.**

Abstract

A tree on n vertices is **graceful** if there exists a bijection $f : V \to \{0, 1, \ldots, n-1\}$ such that the induced edge differences $\{|f(u) - f(v)| : uv \in E\}$ equal $\{1, 2, \ldots, n-1\}$. The graceful tree conjecture (Ringel–Kotzig / Rosa) asserts that every tree is graceful.

On this box we:

1. Enumerate all nonisomorphic trees on $n \leq N = 8$ vertices via `networkx.nonisomorphic_trees` (counts match OEIS A000055).
2. Search for a graceful labeling of each tree by degree-ordered backtracking with used-label / used-difference masks and full end verification.
3. Ledger literature status: conjecture **OPEN**; published exhaustive verification through $n = 35$ (Fang 2010, arXiv:1003.3045) — **not** re-proved here.

Box result: **48**/48 trees graceful (**0** failures; success rate **100%**). Conjecture remains **OPEN**.

Problem statement

Graceful tree conjecture. Every tree admits a graceful labeling.

This experiment. Exact finite search for $n \leq 8$ (delivery-capped; OEIS counts $1, 1, 1, 2, 3, 6, 11, 23$). No infinitary claim is asserted.

Literature ledger (not re-proved on box)

Item	Status	Note
Graceful tree conjecture	**OPEN**	No proof or counterexample
Rosa / Aldred–McKay / Horton	Literature	Historical exhaustive bounds through ~16 / 27 / 28–29
Fang (2010) arXiv:1003.3045	Literature	Every tree with $\leq$ **35** vertices is graceful

Item	Status	Note
Fang (2012) unpublished ≤ 39	Unverified claim	Not treated as published here (MathWorld)
Almost-graceful asymptotics	Literature	Large trees have gracesize $(1-\varepsilon)n$ (e.g. arXiv:2511.11331) — not full GTC
Clay / ToE	**N/A**	Not a Clay problem; unrelated to ToE

One-liner. Graceful tree conjecture remains **OPEN**; published computation verifies all trees through $n = 35$ (Fang); this box only re-checks $n \leq 8$.

Method (box)

1. Generate each nonisomorphic n-vertex tree (`networkx.nonisomorphic_trees`).
2. Backtrack assign labels $\{0, \ldots, n-1\}$ in degree-descending vertex order.
3. Reject partial assignments that repeat an edge difference (including same-step collisions among already-labeled neighbors).
4. Accept only if the completed map induces differences $\{1, \ldots, n-1\}$.
5. Unit tests: path P_3, star $K_{1,3}$ succeed; non-graceful fake map on P_4 rejected; $n = 6$ tree count $= 6$.

Results (this box, 2026-09-11)

n	Trees (OEIS A000055)	Graceful found	Failures	Wall (s)
1	1	1	0	$< 10^{-3}$
2	1	1	0	$< 10^{-3}$
3	1	1	0	$< 10^{-3}$
4	2	2	0	$< 10^{-3}$
5	3	3	0	$< 10^{-3}$
6	6	6	0	$< 10^{-3}$
7	11	11	0	0.001
8	**23**	**23**	**0**	0.008
Total	**48**	**48**	**0**	≈ 0.01

Unit tests: **PASS**. Success rate: **100%** for all checked trees.

Honesty verdict

Finite verification only. Every nonisomorphic tree on $n \leq 8$ vertices admitted a graceful labeling under exhaustive search on this box (**0** failures). Literature extends published exhaustive verification to $n = 35$ (Fang 2010) but does not close the problem. **The graceful tree conjecture remains OPEN.** This note does **not** claim the conjecture proved, and does **not** claim Clay / ToE.

Reproducibility

```
cd /workspace/experiments/band1/graceful_tree
python3 graceful_tree_search.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `README.md`.

References (selected)

1. A. Rosa, *On certain valuations of the vertices of a graph*, Theory of Graphs (Internat. Sympos., Rome, 1966), 1967.
2. R. E. L. Aldred and B. D. McKay, *Graceful and harmonious labelings of trees*, Bull. Inst. Combin. Appl. **23** (1998).
3. W. Fang, *A computational approach to the graceful tree conjecture*, arXiv:1003.3045 (2010).
4. Wolfram MathWorld, *Graceful Tree Theorem* (https://mathworld.wolfram.com/GracefulTreeTheorem.html).